Artificial Intelligence Practical

UNIVERSITY OF DELHI

Ramanujan College

DSC 16: Artificial Intelligence

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Submitted By:

~Ratnesh Kumar

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Submitted To:

~ Ms. Bhavya Ahuja Grover

Acknowledgement

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At last but not least, I am very thankful to God almighty for showering his blessings upon me.

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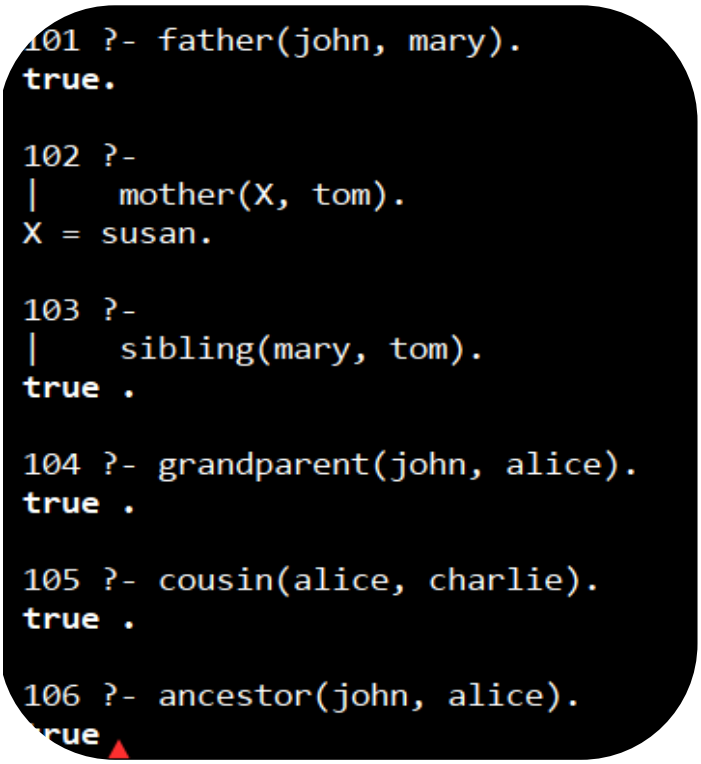
1.Write a PROLOG program to implement the family tree and demonstrate the family relationship.

Code:

```
parent(john, mary).

parent(john, tom).

parent(john, ann).

parent(susan, mary).

parent(susan, tom).

parent(susan, ann).

parent(mary, alice).

parent(mary, bob).

parent(tom, charlie).

parent(tom, diana).

parent(ann, edward).

male(john).

male(tom).

male(bob).

male(charlie).

male(edward).

female(susan).

female(mary).

female(ann).

female(alice).

female(diana).

father(X, Y) :- parent(X, Y), male(X).

mother(X, Y) :- parent(X, Y), female(X).

child(X, Y) :- parent(Y, X).

son(X, Y) :- child(X, Y), male(X).

daughter(X, Y) :- child(X, Y), female(X).

sibling(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

brother(X, Y) :- sibling(X, Y), male(X).

sister(X, Y) :- sibling(X, Y), female(X).

grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

grandfather(X, Y) :- grandparent(X, Y), male(X).

grandmother(X, Y) :- grandparent(X, Y), female(X).

grandchild(X, Y) :- grandparent(Y, X).

grandson(X, Y) :- grandchild(X, Y), male(X).

granddaughter(X, Y) :- grandchild(X, Y), female(X).

uncle(X, Y) :- brother(X, Z), parent(Z, Y).

aunt(X, Y) :- sister(X, Z), parent(Z, Y).

cousin(X, Y) :- parent(Z, X), parent(W, Y), sibling(Z, W).

ancestor(X, Y) :- parent(X, Y).

ancestor(X, Y) :- parent(X, Z), ancestor(Z, Y).

descendant(X, Y) :- ancestor(Y, X).
```

Output:

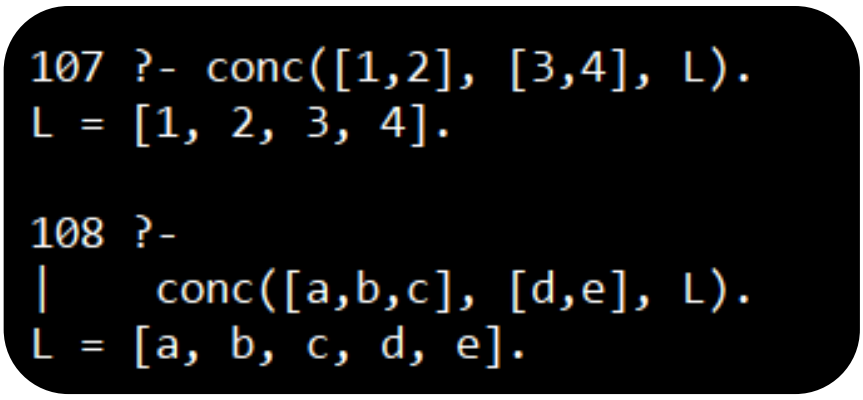


2. Write a PROLOG program to implement conc(L1, L2, L3) where L2 is the list to be appended with L1 to get the resulted list L3.

Code:

```
conc([], L2, L2).

conc([H|T1], L2, [H|T3]) :- conc(T1, L2, T3).
```

Output:

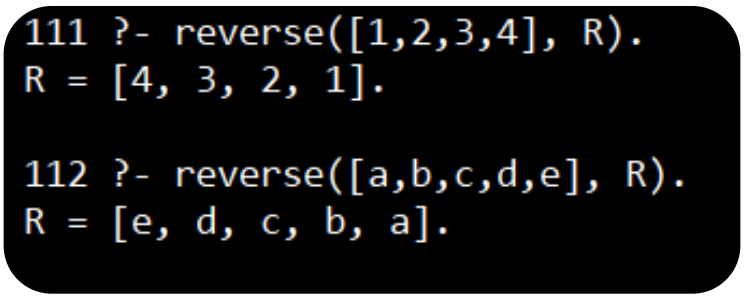


3. Write a PROLOG program to implement reverse(L, R) where List L is original and List R is reversed list.

Code:

```
reverse_list(L, R):-
    rev(L,[],R).
rev([], Acc, Acc).
rev([H|T],Acc, R) :-
    rev(T, [H|Acc], R).
```

Output:

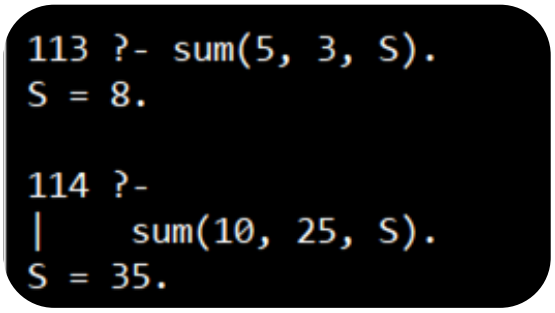


4. Write a PROLOG program to calculate the sum of two numbers.

Code:

```
sum(X, Y, Sum) :- Sum is X + Y.
```

Output:

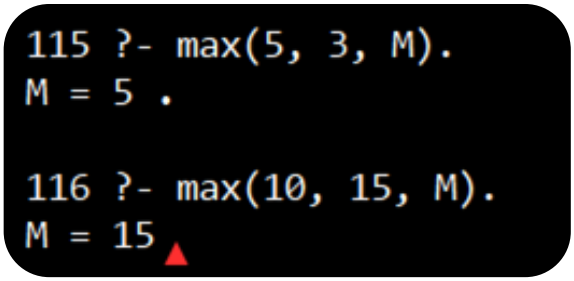


5. Write a PROLOG program to implement max(X, Y, M) so that M is the maximum of two numbers X and Y.

Code:

```
max(X, Y, X) :- X >= Y.

max(X, Y, Y) :- Y > X.
```

Output:

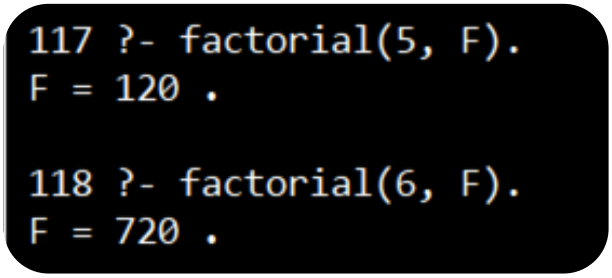


6. Write a program in PROLOG to implement factorial (N, F) where F represents the factorial of a number N.

Code:

```
factorial(0, 1).

factorial(N, F) :-
    N > 0,
    N1 is N - 1,
    factorial(N1, F1),
    F is N * F1.
```

Output:

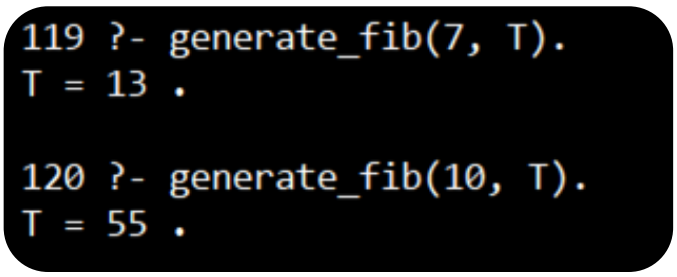


7. Write a program in PROLOG to implement generate_fib(N,T) where T represents the Nth term of the Fibonacci series.

Code:

```
generate_fib(0, 0).

generate_fib(1, 1).

generate_fib(N, T) :-

    N > 1,

    N1 is N - 1,

    N2 is N - 2,

    generate_fib(N1, T1),

    generate_fib(N2, T2),

    T is T1 + T2.
```

Output:

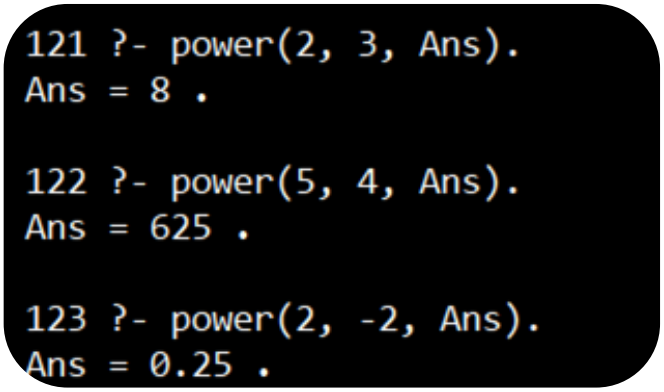


8. Write a PROLOG program to implement power (Num, Pow, Ans) : where Num is raised to the power Pow to get Ans.

Code:

```
power(_, 0, 1).
power(Num, Pow, Ans) :-
    Pow > 0,
    Pow1 is Pow - 1,
    power(Num, Pow1, Ans1),
    Ans is Num * Ans1.
power(Num, Pow, Ans) :-
    Pow < 0,
    PosPow is -Pow,
    power(Num, PosPow, TempAns),
    Ans is 1 / TempAns.
```

Output:

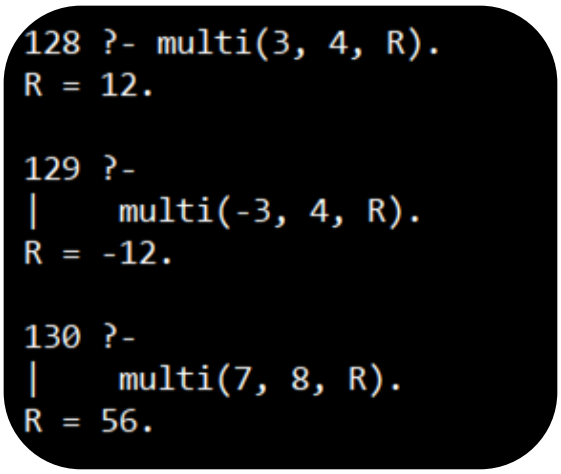


9. PROLOG program to implement multi (N1, N2, R) : where N1 and N2 denotes the numbers to be multiplied and R represents the result.

Code:

```
multi (N1, N2, R) :- R is N1 * N2.
```

Output:

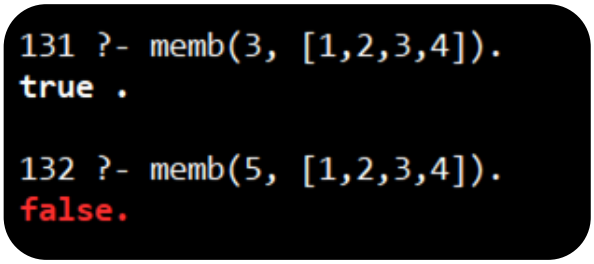


10. Write a PROLOG program to implement memb(X, L): to check whether X is a member of L or not.

Code:

```
memb(X, [X|_]).

memb(X, [_|T]) :- memb(X, T).
```

Output:



11. Write a PROLOG program to implement sumlist(L, S) so that S is the sum of a given list L.

Code:

```
sumlist([], 0).

sumlist([H|T], S) :-

    sumlist(T, S1),

    S is H + S1.
```

Output:

```
133 ?- sumlist([1,2,3,4,5], S).
S = 15.

134 ?-
|    sumlist([10,20,30], S).
S = 60.
```

12. Write a PROLOG program to implement two predicates evenlength(List) and oddlength(List) so that they are true if their argument is a list of even or odd length respectively.

Code:

```
evenlength([]).

evenlength([_|T]) :- oddlength(T).

oddlength([_]).

oddlength([_|T]) :- evenlength(T).
```

Output:

```
135 ?- evenlength([1,2,3,4]).
true .

136 ?- oddlength([1,2,3]).
true .

137 ?- evenlength([a,b]).
true .
```

13. Write a PROLOG program to implement maxlist(L, M) so that M is the maximum number in the list.

Code:

```
max(X, Y, X) :- X >= Y.

max(X, Y, Y) :- Y > X.

maxlist([X], X).

maxlist([H|T], M) :-
    maxlist(T, M1),
    max(H, M1, M).
```

Output:

```
138 ?- maxlist([3,7,2,9,1], M).
M = 9 .

139 ?- maxlist([5,2,8,1,9,3], M).
M = 9 .
```

14. Write a PROLOG program to implement insert(I, N, L, R) that inserts an item I into Nth position of list L to generate a list R.

Code:

```
insert(I, 1, L, [I|L]).

insert(I, N, [H|T], [H|R]) :-
    N > 1,
    N1 is N - 1,
    insert(I, N1, T, R).
```

Output:

```
140 ?- insert(x, 3, [a,b,c,d], R).
R = [a, b, x, c, d] .

141 ?- insert(5, 1, [1,2,3], R).
R = [5, 1, 2, 3] .
```

15. Write a PROLOG program to implement delete(N, L, R) that removes the element on Nth position from a list L to generate a list R.

Code:

```
delete(1, [_|T], T).
delete(N, [H|T], [H|R]) :-
    N > 1,
    N1 is N - 1,
    delete(N1, T, R).
```

Output:

```
142 ?- delete(2, [a,b,c,d], R).
R = [a, c, d] .

143 ?- delete(1, [1,2,3,4], R).
R = [2, 3, 4] .
```